



The Process Orchestration Journey

Getting from 0 to 100 with process automation

CamundaCon 2022
Daniel Meyer & Bernd Ruecker



“
We need to speed up
our bank account
opening. **Others do
this in minutes, we
need 3 days!**



The as-is situation

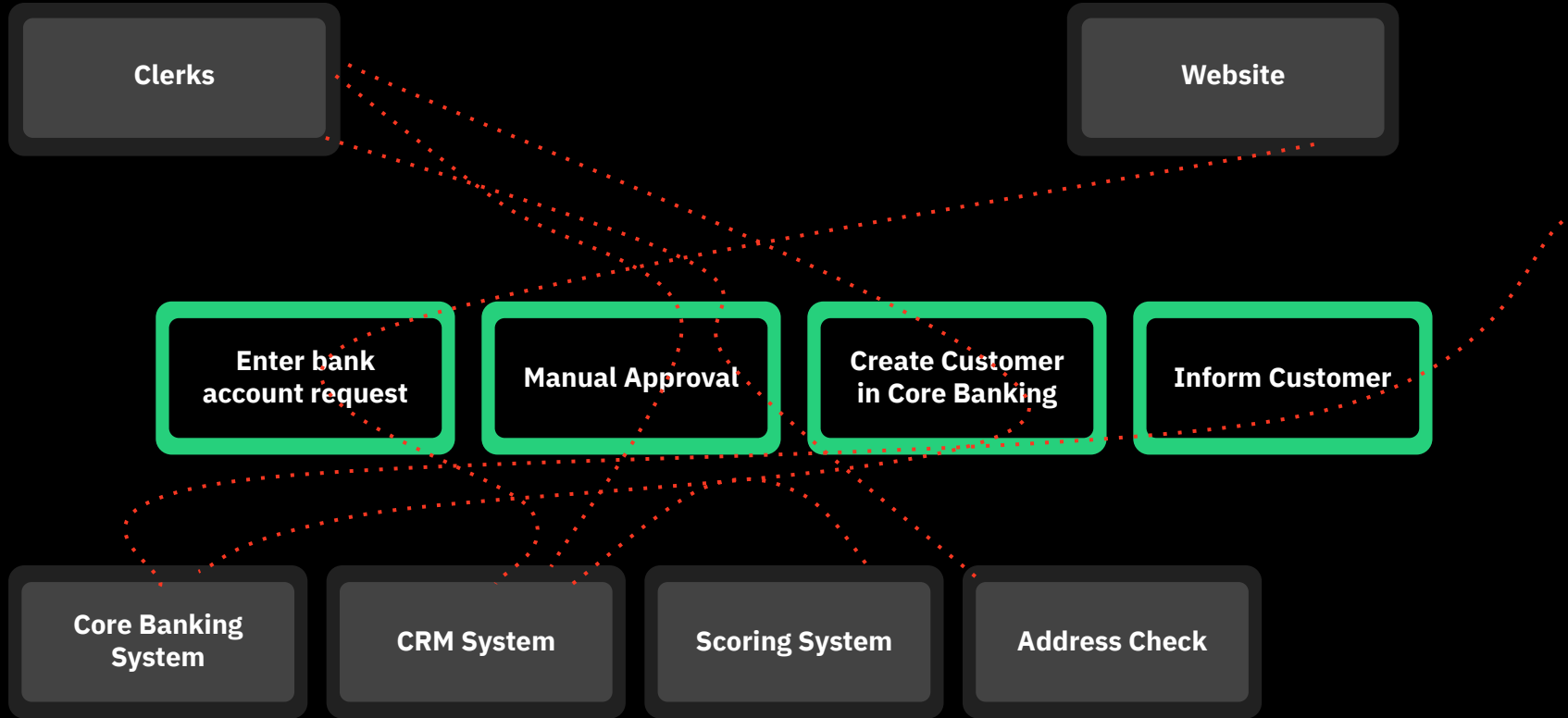
**Enter bank
account request**

Manual Approval

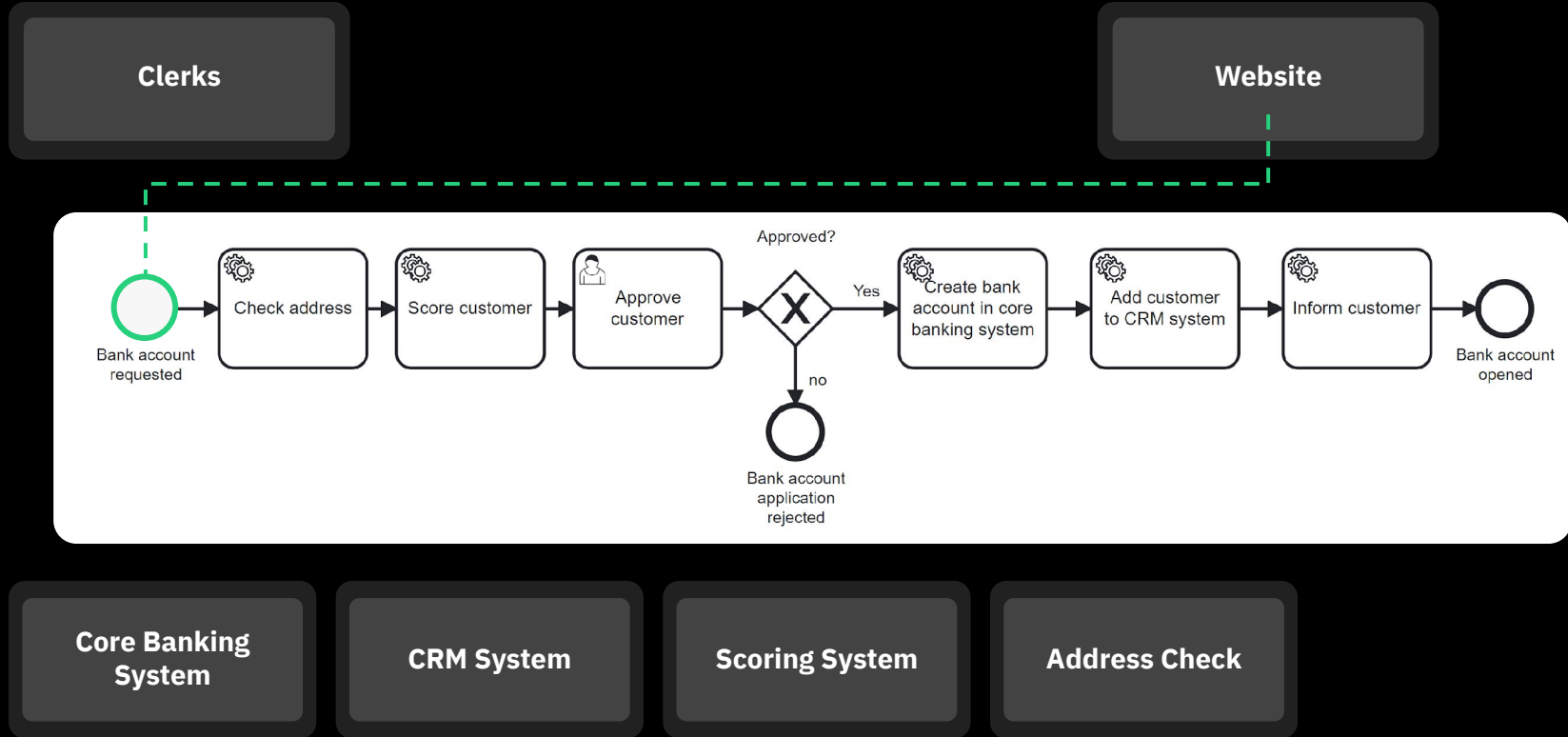
**Create Customer
in Core Banking**

Inform Customer

The as-is situation



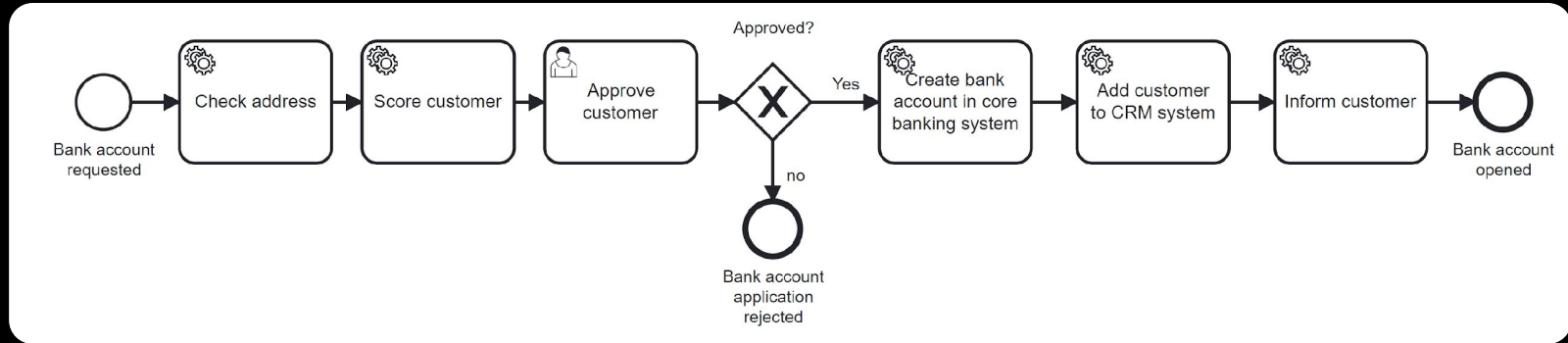
A first automation project



A first automation project

Clerks

Website



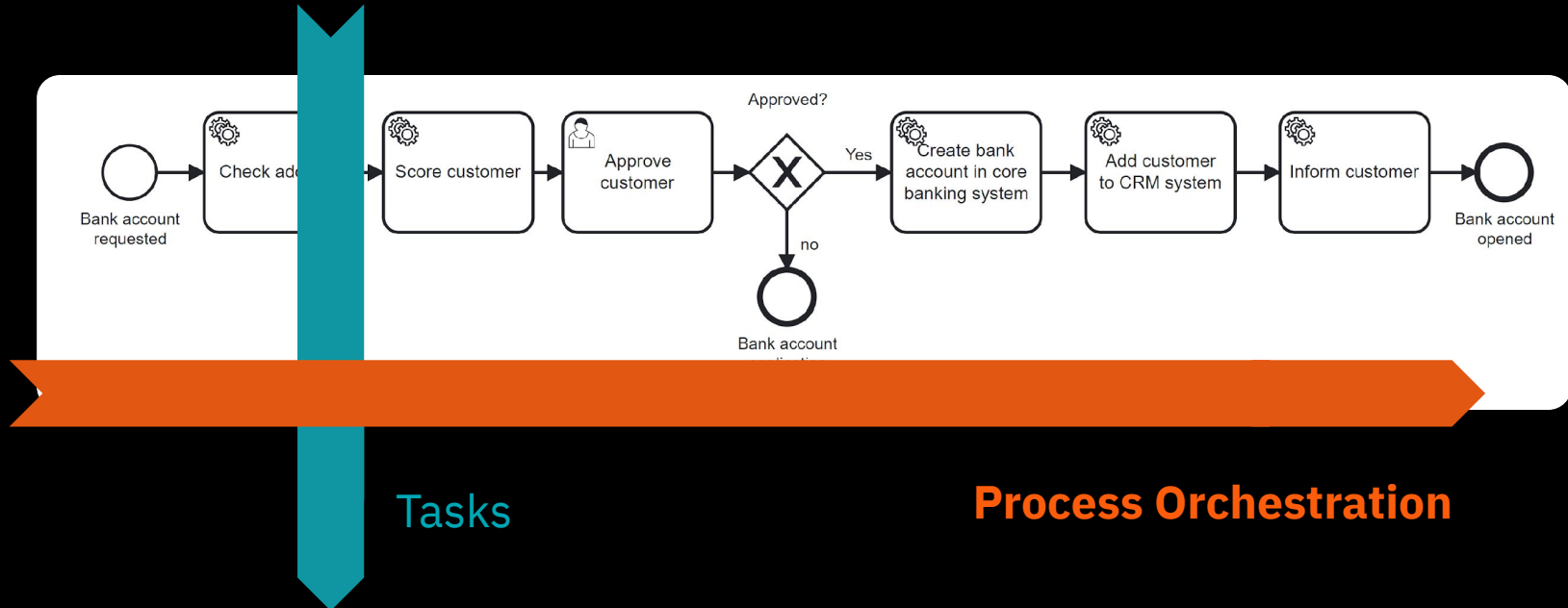
Core Banking
System

CRM System

Scoring System

Address Check

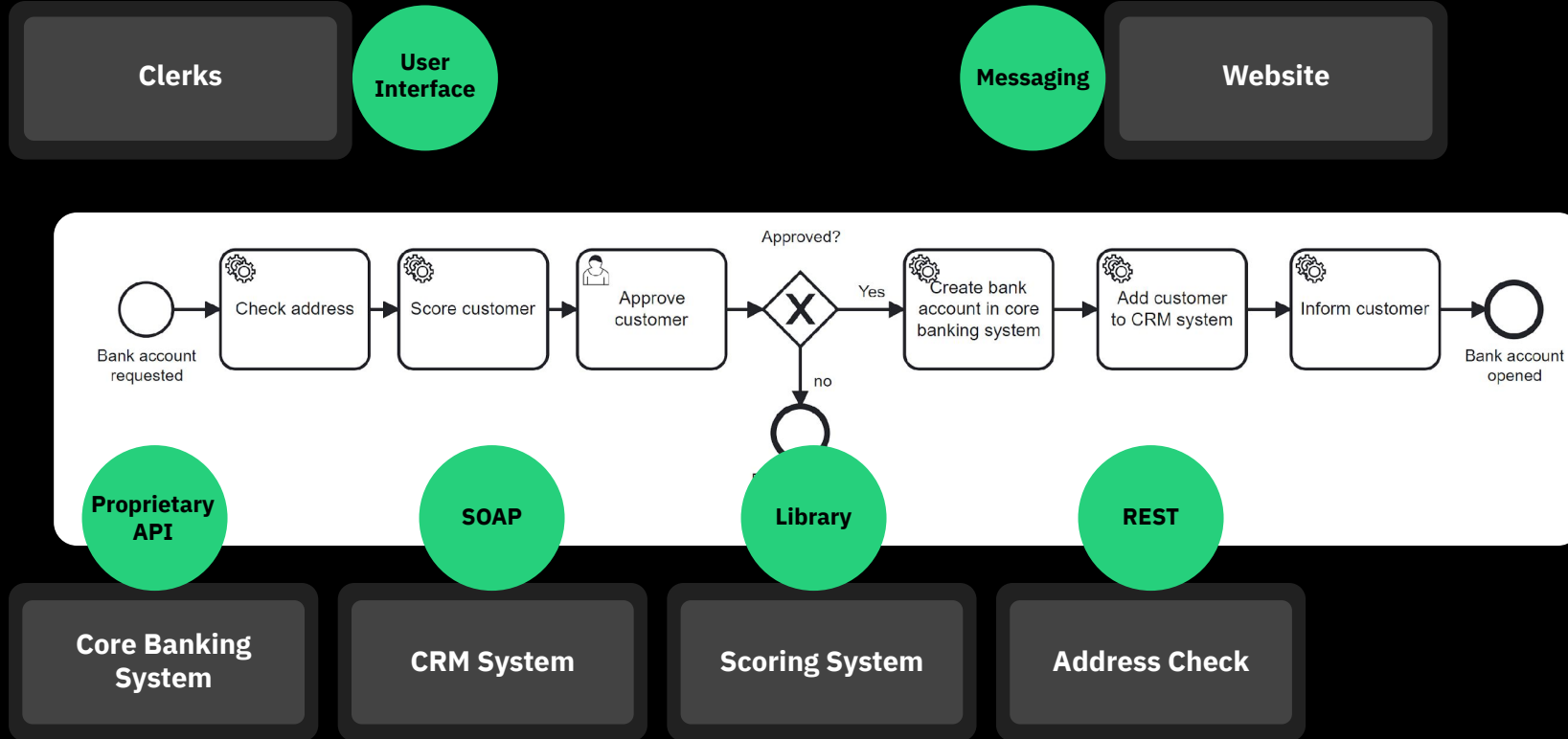
Task vs. process automation



Tasks

Process Orchestration

Endpoint diversity



Endpoint diversity



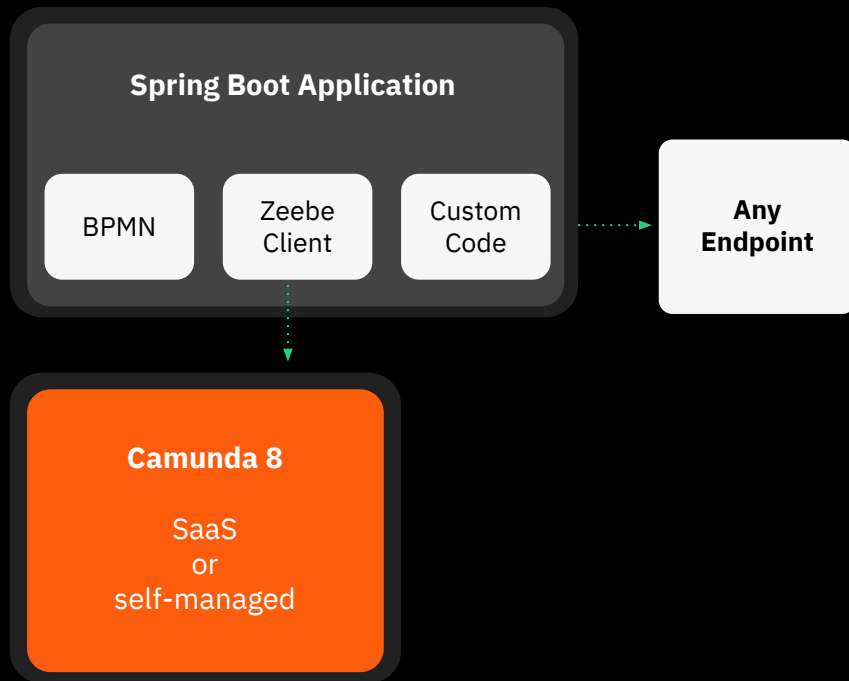
No problem!
Just code
in Spring
Boot!

Demo

<https://github.com/berndruecker/customer-onboarding-camunda-8-springboot/tree/camunda-con-2022>



Solution architecture



Supports any programming language

- Clients for Java, C#, NodeJS, Go, ... available
- Natively integrates into your stack

Runs anywhere

- Available as a SaaS service
- Manage it yourself (in your own cloud, your datacenter, your laptop)

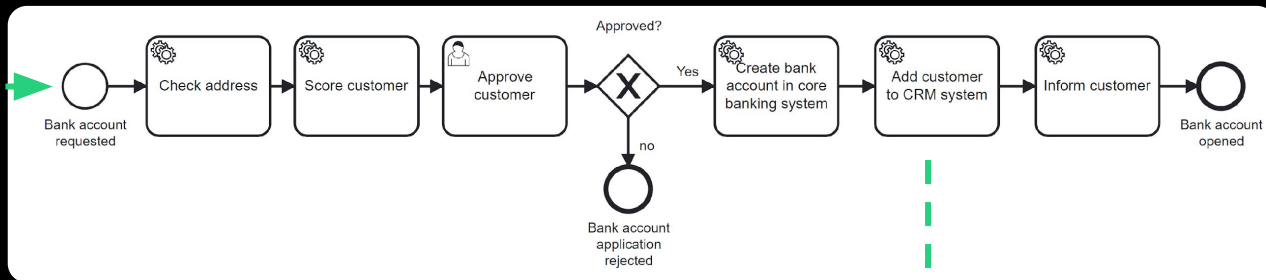
```

@PutMapping("/customer")
public ResponseEntity<CustomerOnboardingResponse> onboardCustomer(ServerWebExchange exchange) {
    HashMap<String, Object> variables = new HashMap<>();
    variables.put("automaticProcessing", true);
    variables.put("someInput", "yeah");

    client.newCreateInstanceCommand() //
        .bpmnProcessId("customer-onboarding") //
        .latestVersion() //
        .variables(variables) //
        .send().join();
    return ResponseEntity.status(HttpStatus.ACCEPTED).build();
}

```

Your code to provide a REST endpoint



Your code to implement the REST call

```

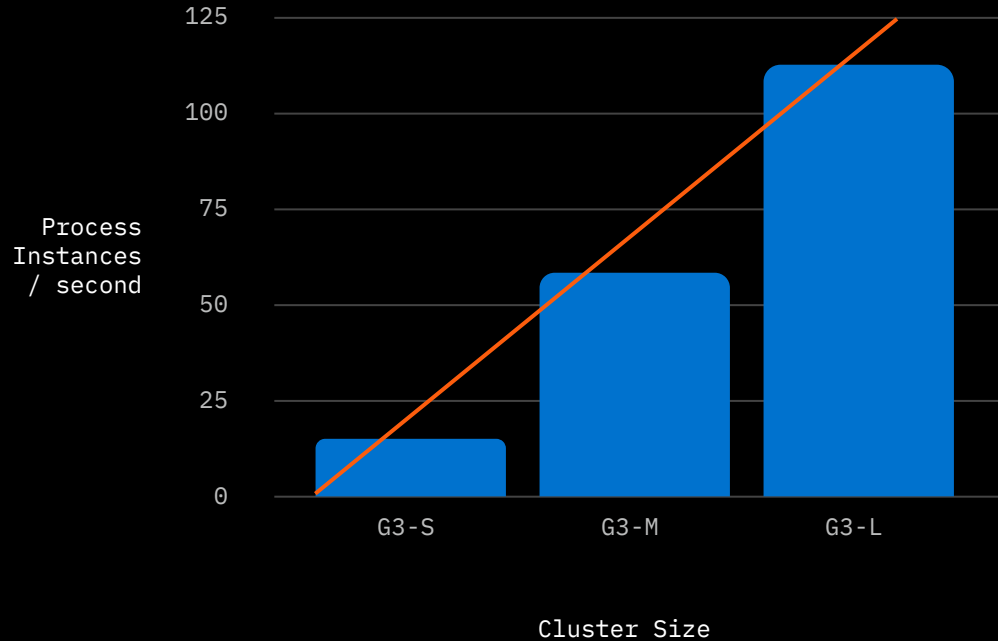
@ZeebeWorker(type = "addCustomerToCrm", autoComplete = true)
public void addCustomerToCrmViaREST(final ActivatedJob job) throws IOException {
    String request = "someData";
    restTemplate.put(ENDPOINT_CRM, request);
}

```

Scalability and resiliency built-in

Camunda 8

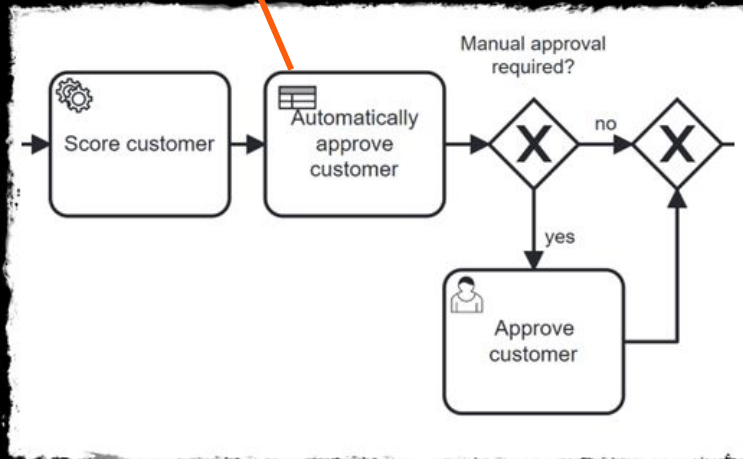
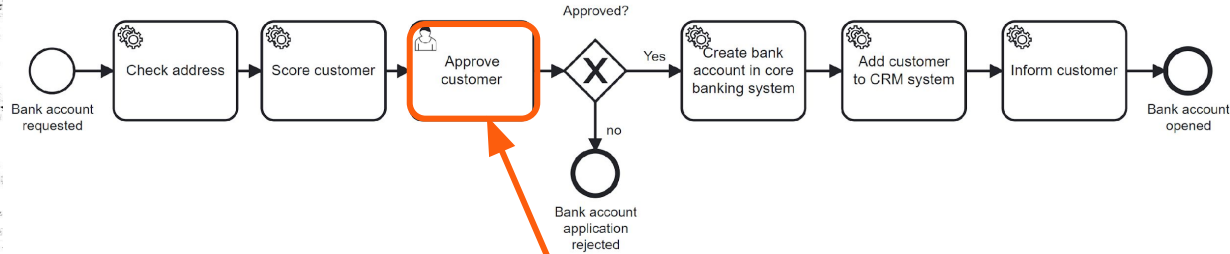
SaaS
or
self-managed



“We need to speed up our bank account opening. Others do this **in minutes**, we need 3 days!



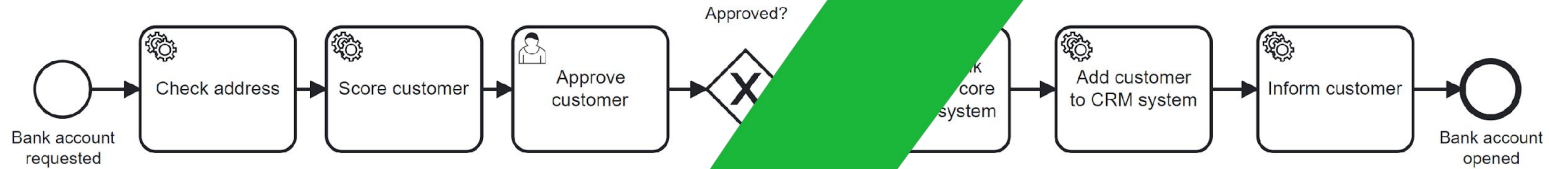
Changing tasks



A first automation project

Clerks

Website



Core Banking
System

Scoring System

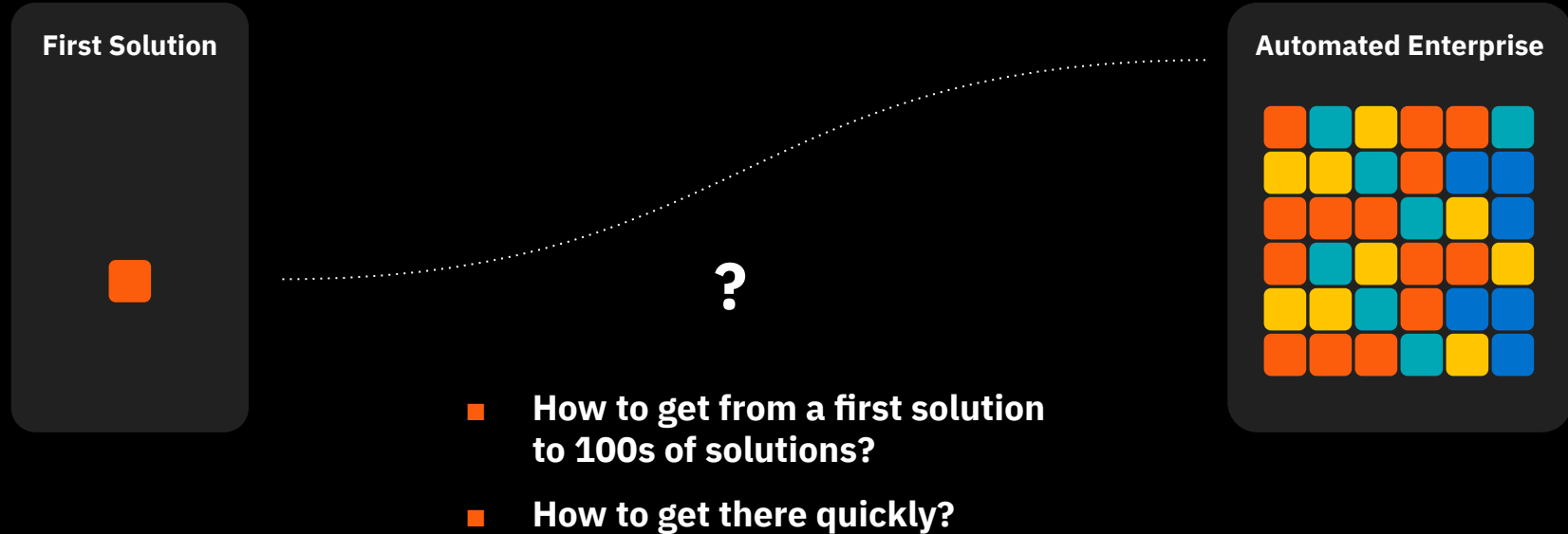
Address Check

Are You the Amazon of your Industry now?



DANIEL

The Transformation Journey

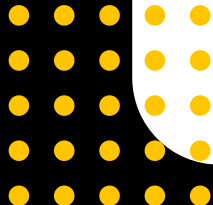


“

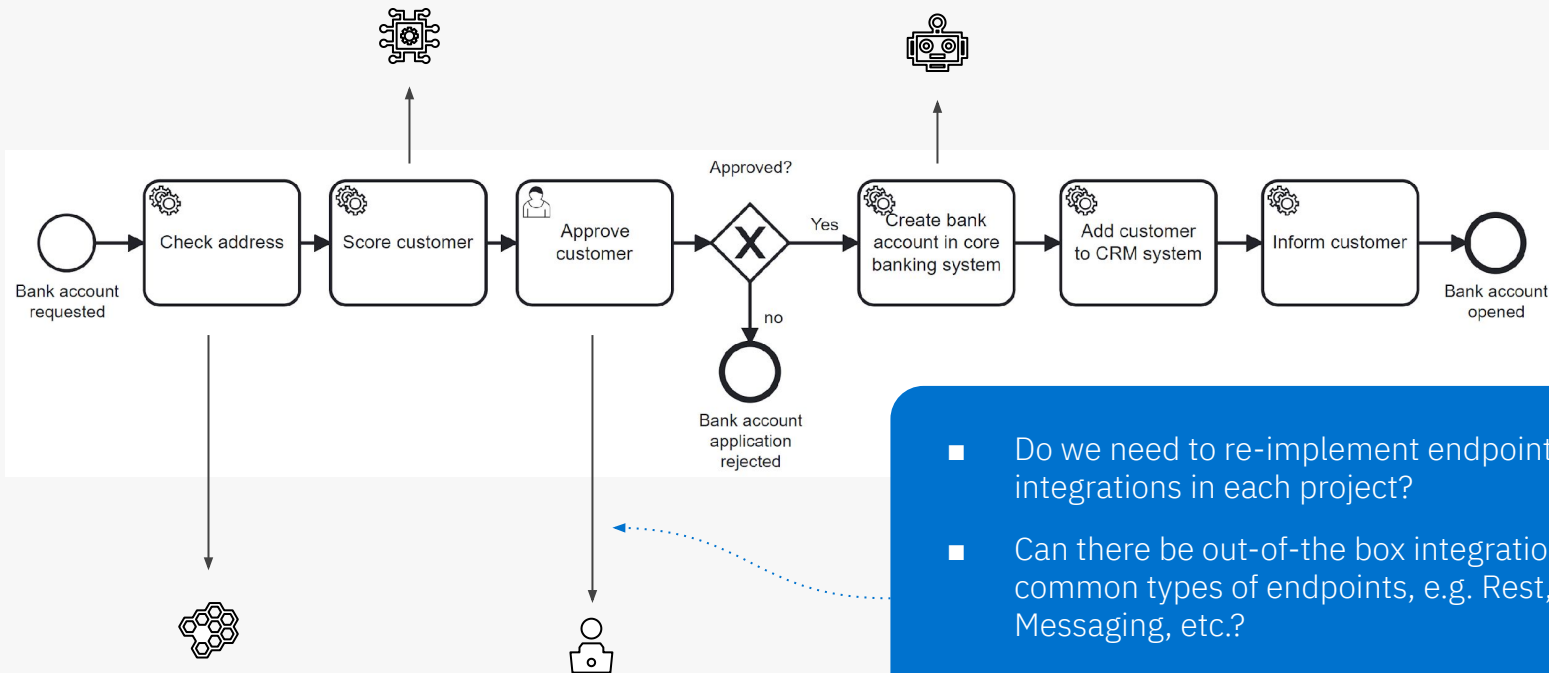
Our vision is to enable a broader range of personas to create and run process orchestration solutions a lot faster than they can today.

Mo Ladha

VP Product Management, Camunda



Let's start by looking at endpoint connectivity

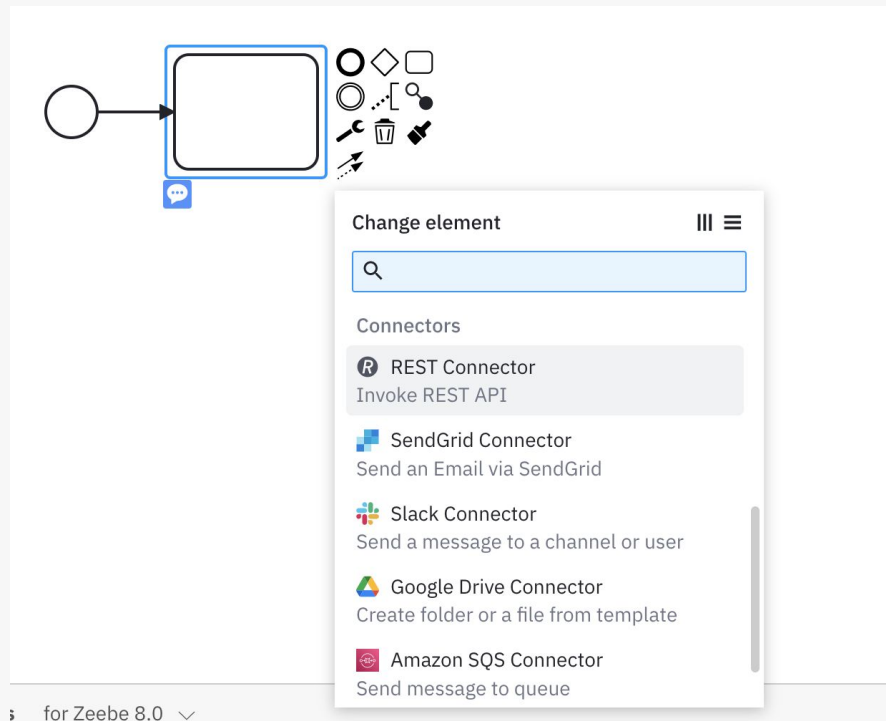


- Do we need to re-implement endpoint integrations in each project?
- Can there be out-of-the box integrations with common types of endpoints, e.g. Rest, Messaging, etc.?
- Can we enable less technical developers to reuse integrations that others have built?

Available out of the box Connectors



- **Amazon SQS Connector** - Send messages to Amazon Simple Queue Service from your BPMN process.
- **AWS Lambda Connector** - Invoke AWS Lambda Functions from your BPMN process.
- **Google Drive Connector** - Create folders or files from a Google Drive template from your BPMN process.
- **REST Connector** - Make a request to a REST API and use the response in the next steps of your process.
- **SendGrid Connector** - Quickly send emails from your BPMN processes.
- **Slack Connector** - Send messages to channels or users in your Slack workspace from your BPMN process.



On the Roadmap (Q4)



Kafka

Send message to
queue



RabbitMQ

Send message to
queue



Amazon SNS

Send message to
topic



Teams

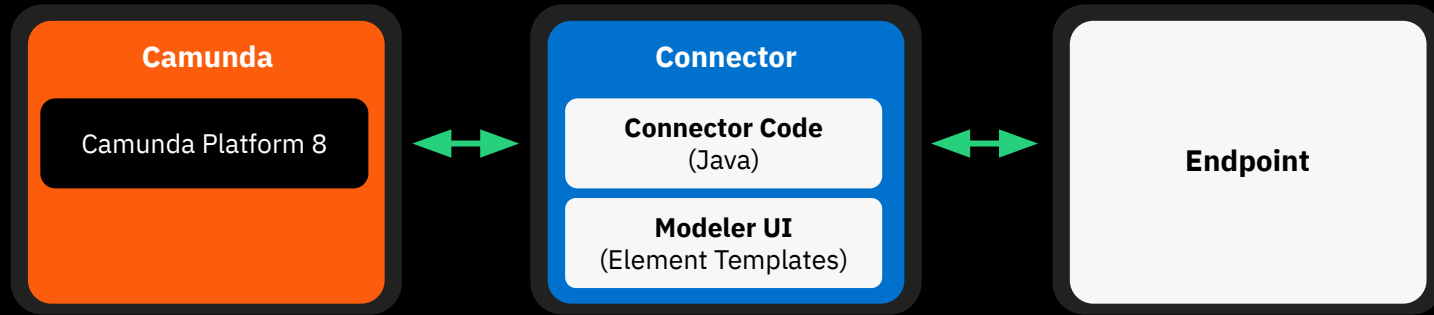
Send message to
channel or user



Sharepoint

Create folder
Update contents of file

What is a Connector?

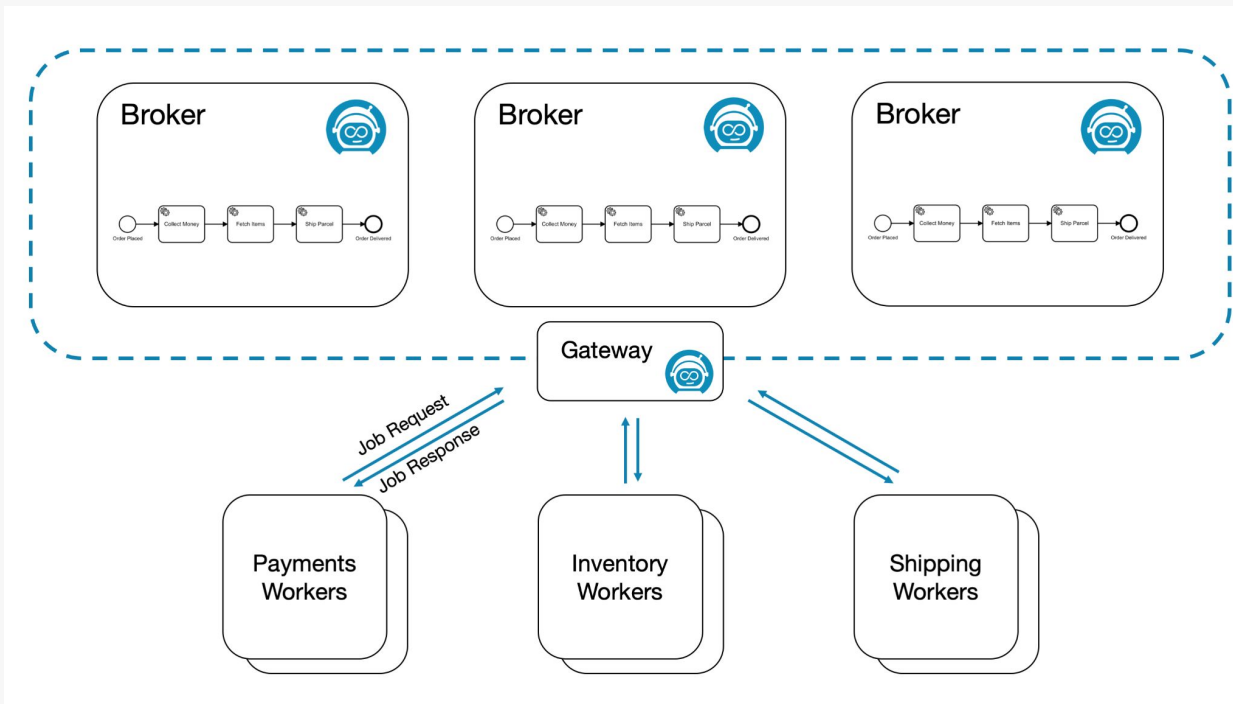


BERND

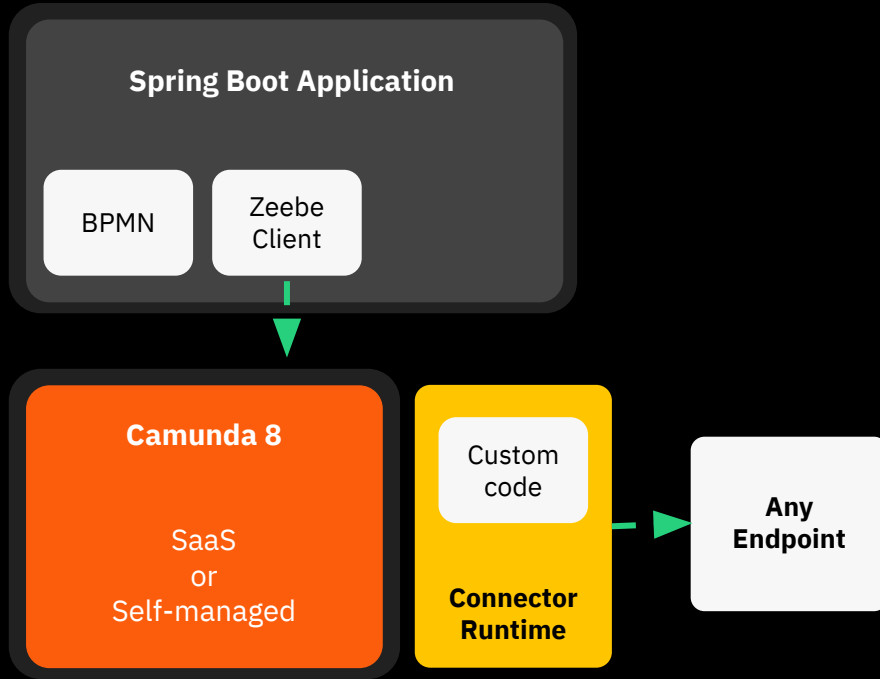
Live Hacking



Camunda Superpower: External Job Workers



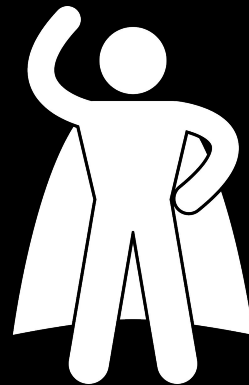
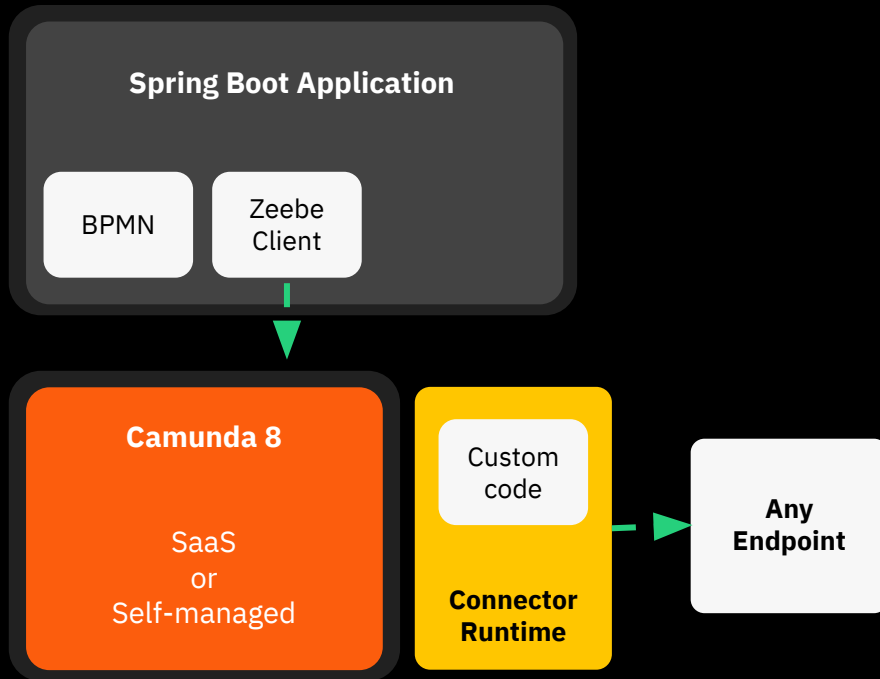
Solution architecture



Runs anywhere

- Multiple connector runtime options, including SaaS, self-managed, and local installations

Solution architecture



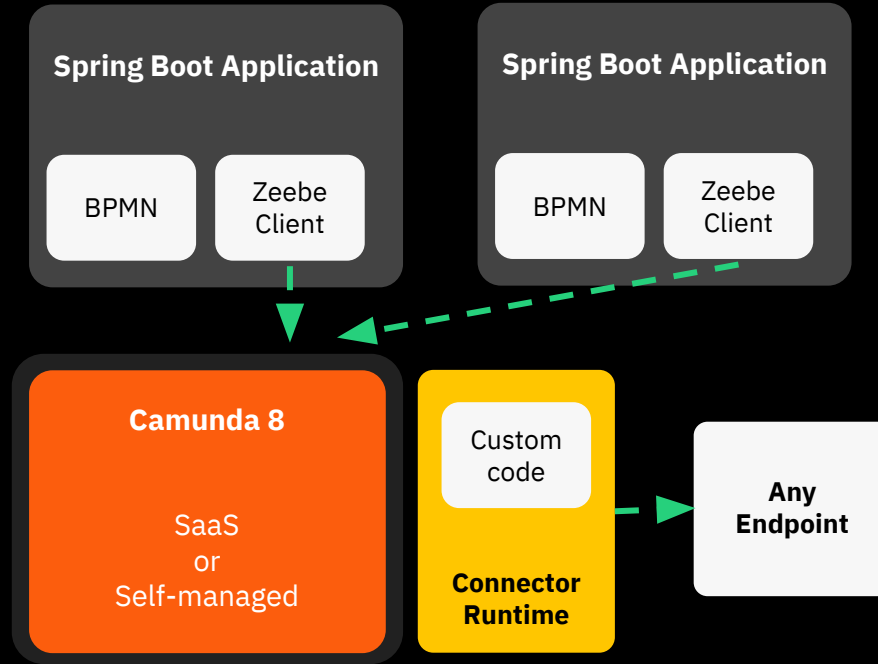
Superpower inside:

External Job Workers

Runs anywhere

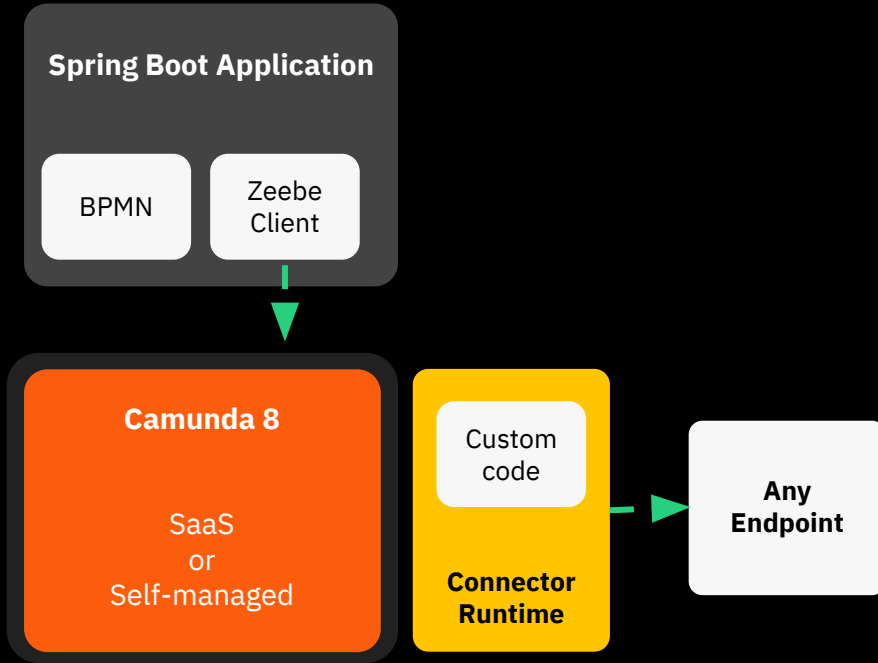
- Multiple connector runtime options, including SaaS, self-managed, and local installations

Scaling connectors



E.g. **Customer onboarding** and **Credit card replacement** can share connectors

Scaling connectors



Multiple connector types

- Outbound
- Inbound
 - Webhook
 - Polling
 - Subscription

Connectivity Vision

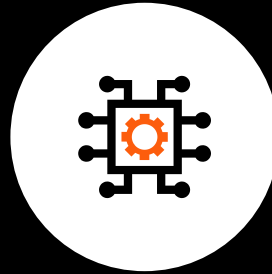


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Connectivity Vision



**Out of the box
connectors**



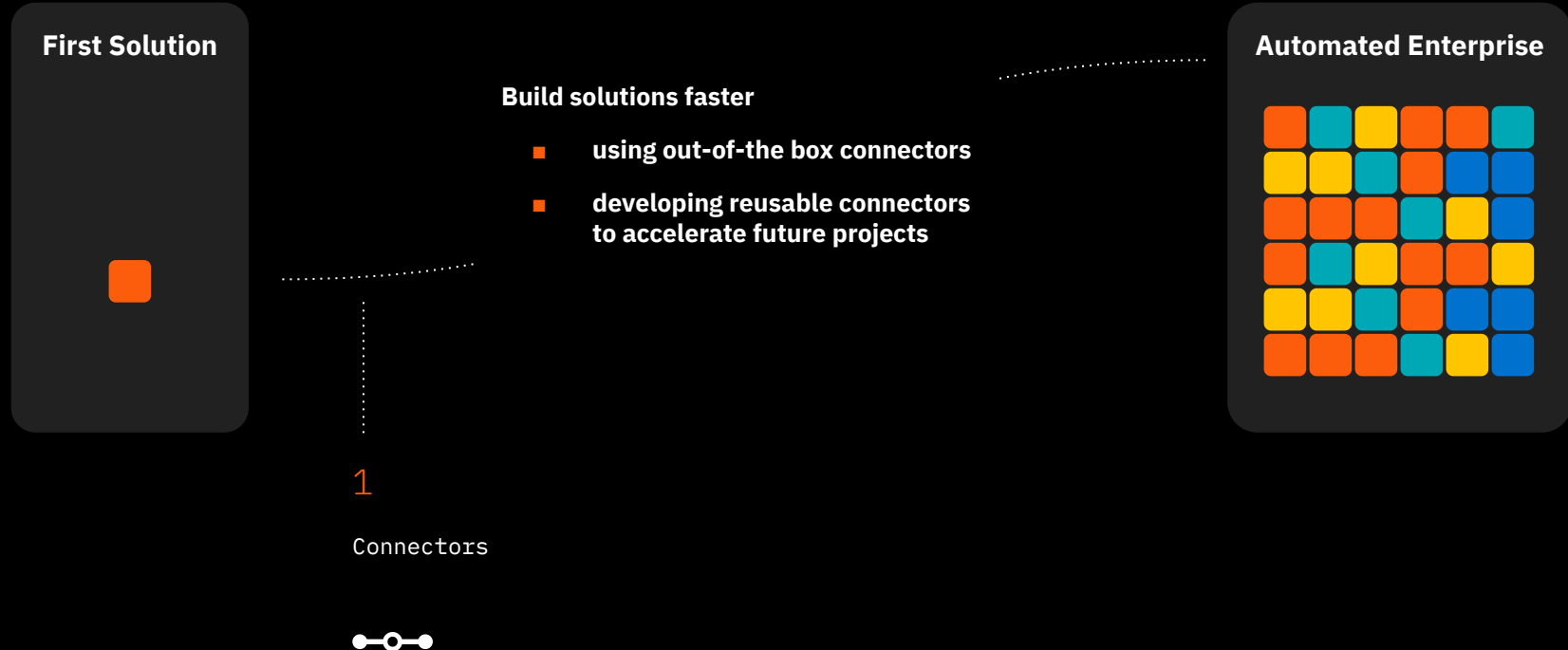
**Connector
SDK**



**Connector
Ecosystem**

The First Accelerator:

Connectors

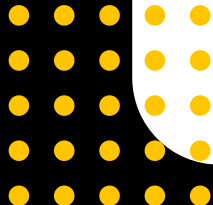


“

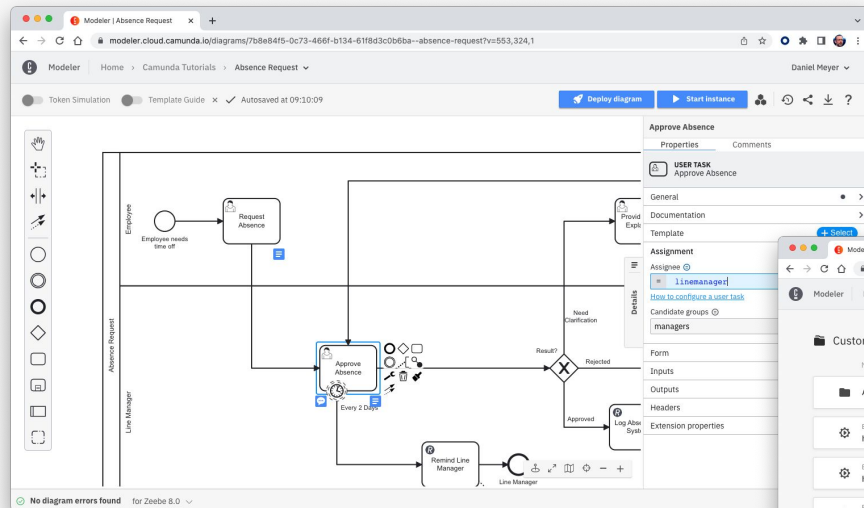
I want to enable any developer, regardless how technical they are, to model & run processes within minutes, not days

Nico Rehwaldt

Creator of Bpmn.io, Camunda



Web Modeler

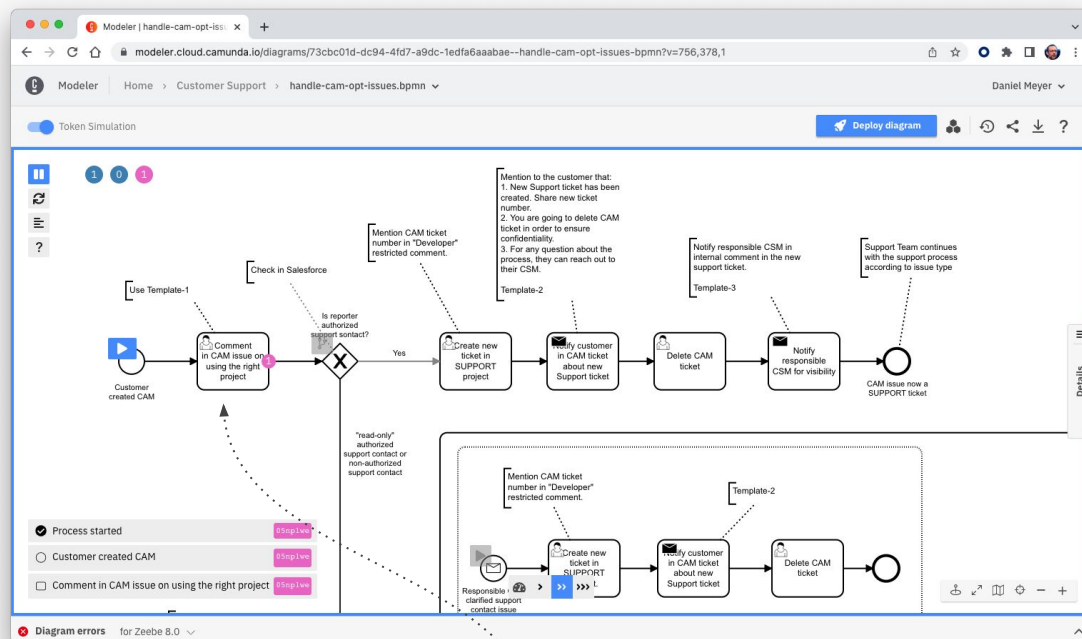


Name	Creator	Last changed
Archive	Garima Yadav	
BPMN diagram handle-cam-opt-issues.bpmn	Garima Yadav	2022-09-21 08:35
BPMN diagram help-request.bpmn	Alexandra Corremans	2022-09-09 19:42
BPMN diagram bug-report.bpmn	Alexandra Corremans	2022-09-08 03:11
BPMN diagram bug-report-new.bpmn	Alexandra Corremans	2022-09-01 07:14
BPMN diagram cc-incident-process.bpmn	You	2022-08-22 15:23
BPMN diagram new-ticket.bpmn	Garima Yadav	2022-08-22 07:47
BPMN diagram support-hiring-process.bpmn	Garima Yadav	2022-08-19 08:37
BPMN diagram	Garima Yadav	2022-08-17 20:40

Name	Role
Alexandra Corremans	Project Admin
Garima Yadav	Project Admin
Jeffrey Stokes	Project Admin
Daniel Ewing	Editor
You	Editor
Lewis Porter	Editor
Michal Dylko	Editor
Neal Dennis	Editor
Renz Patriarca	Editor
Scott Ghidui	Editor

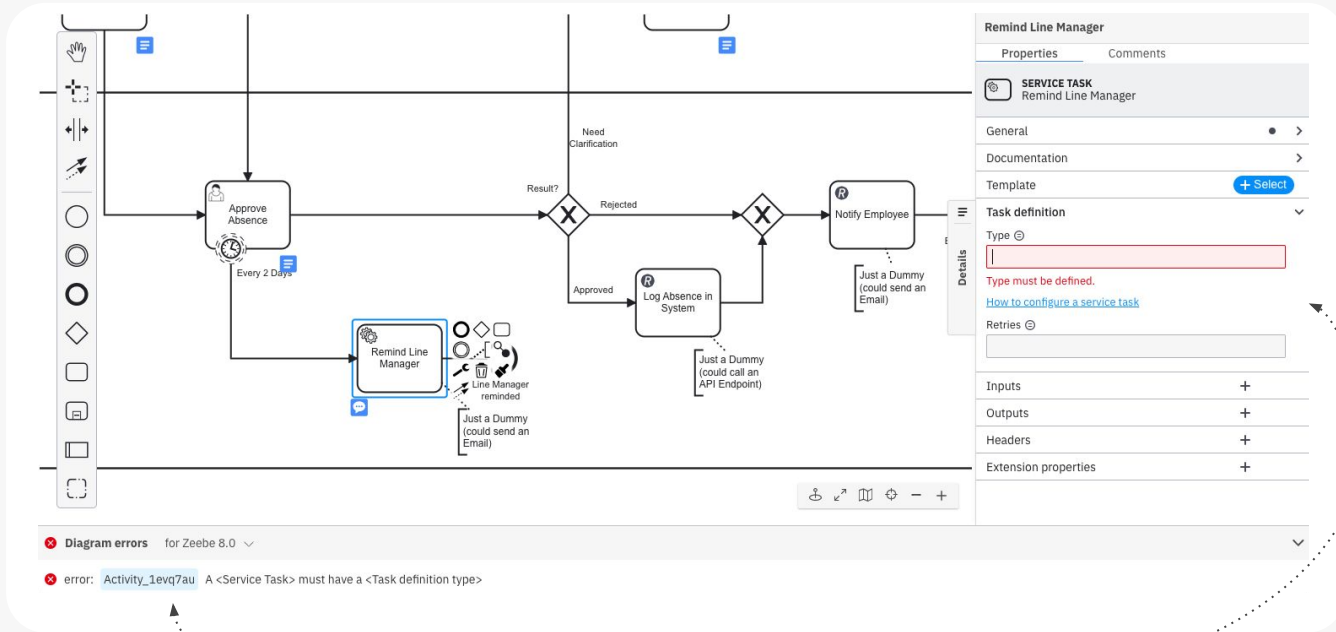
- Shared repository for processes, decisions, forms and templates
- No local installation needed
- Fast onboarding of new developers

Token Simulation



Learn BPMN semantics

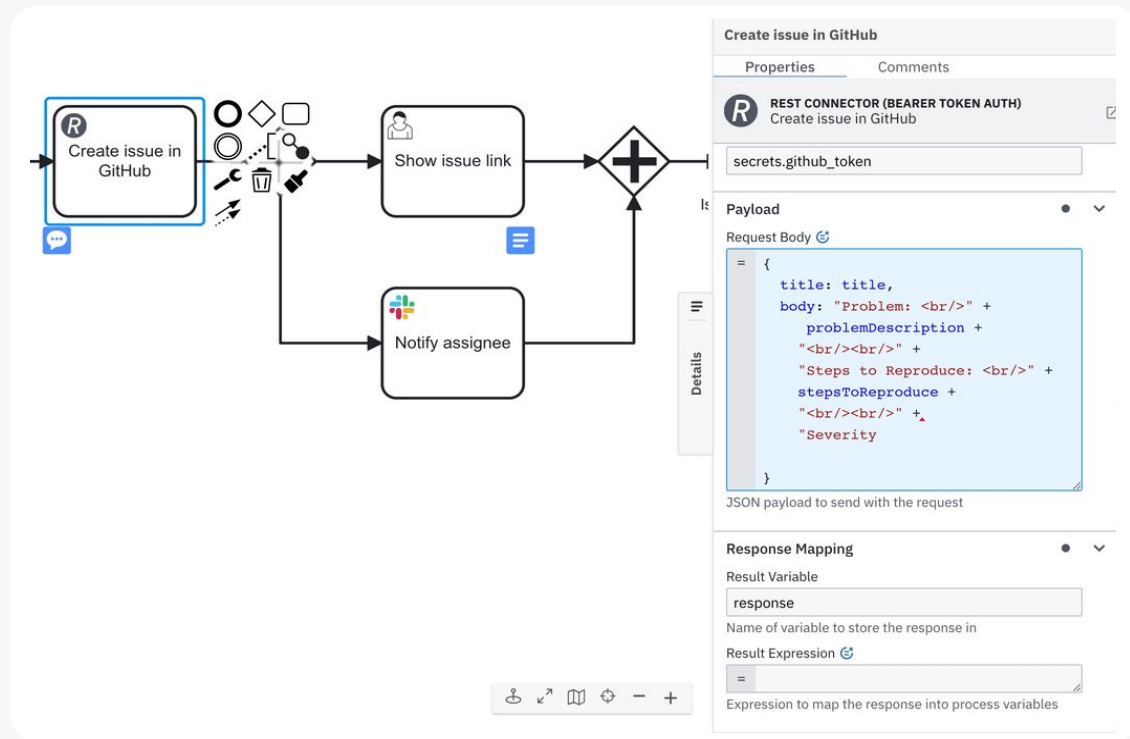
Diagram validation



Improve quality of your diagrams

Prevent errors BEFORE you deploy.

FEEL Editor



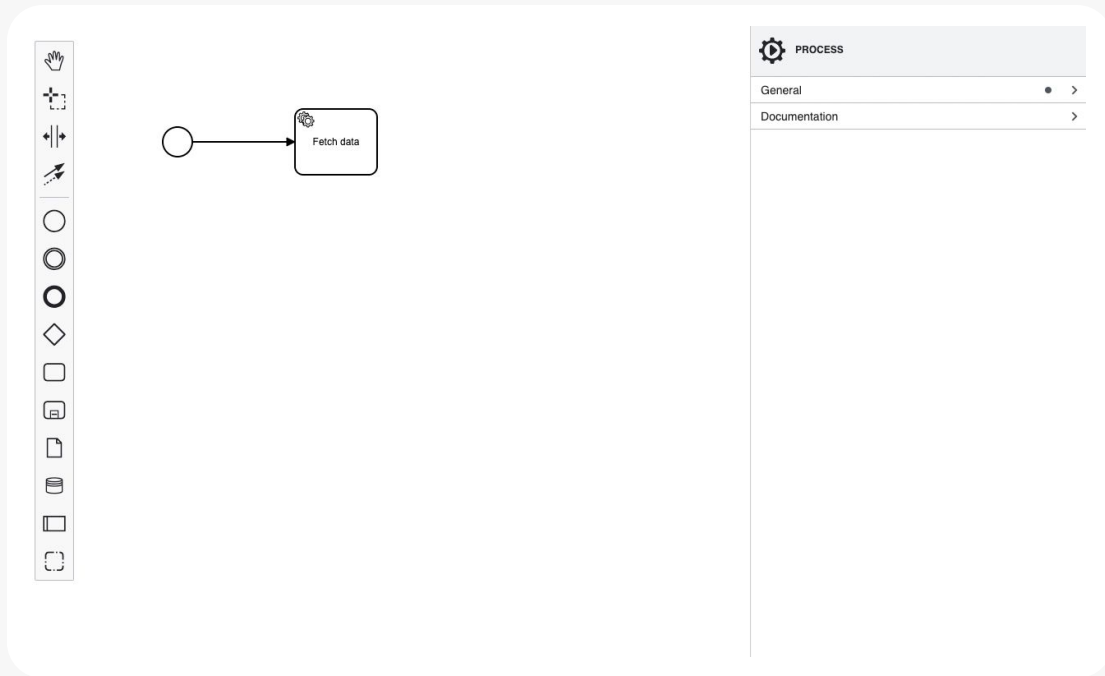
- Syntax validation
- Built-in function suggestion
- Suggesting variable names declared in the process

Coming soon: Data Schemas



Increases productivity, prevents errors, and allow less technical developers to participate in modeling:

- Schema can be defined once, used in every process
- Offers type support



The next Idea: Interactive Debugging



Zeebe Play ▶ Deployment Monitoring Testing

Processes / [solos-transport-process](#) / 2251799813685697

You may need to reload the page manually ↻

{ } Set Variables ▼

Process Instance Key

2251799813685697

Start Time

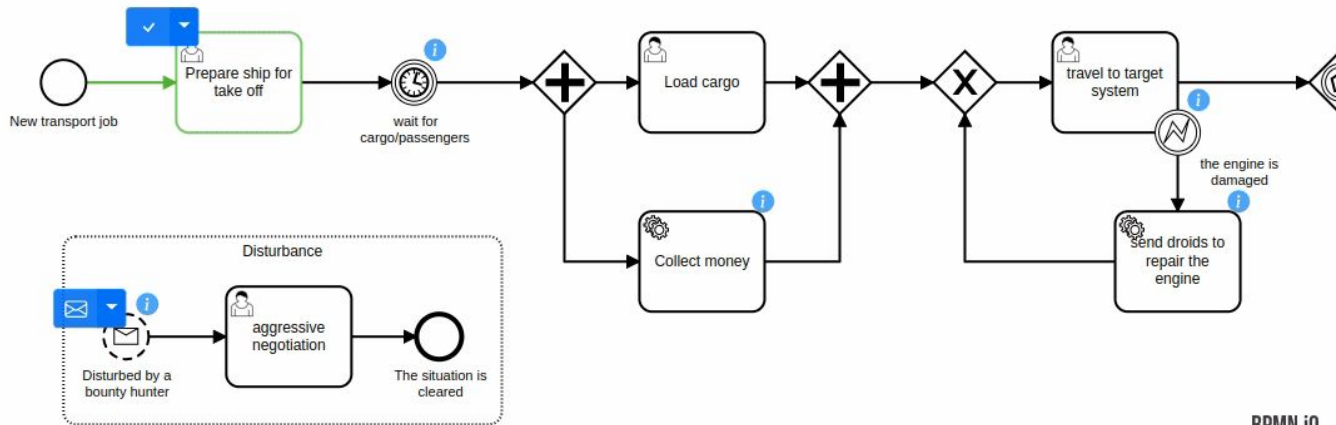
2022-09-04T13:06:58.284+02:00

End Time

-

State

active



BPMN.IO

Variables 3

Element Instances 4

Jobs 1

Incidents 0

Message Subscriptions 1

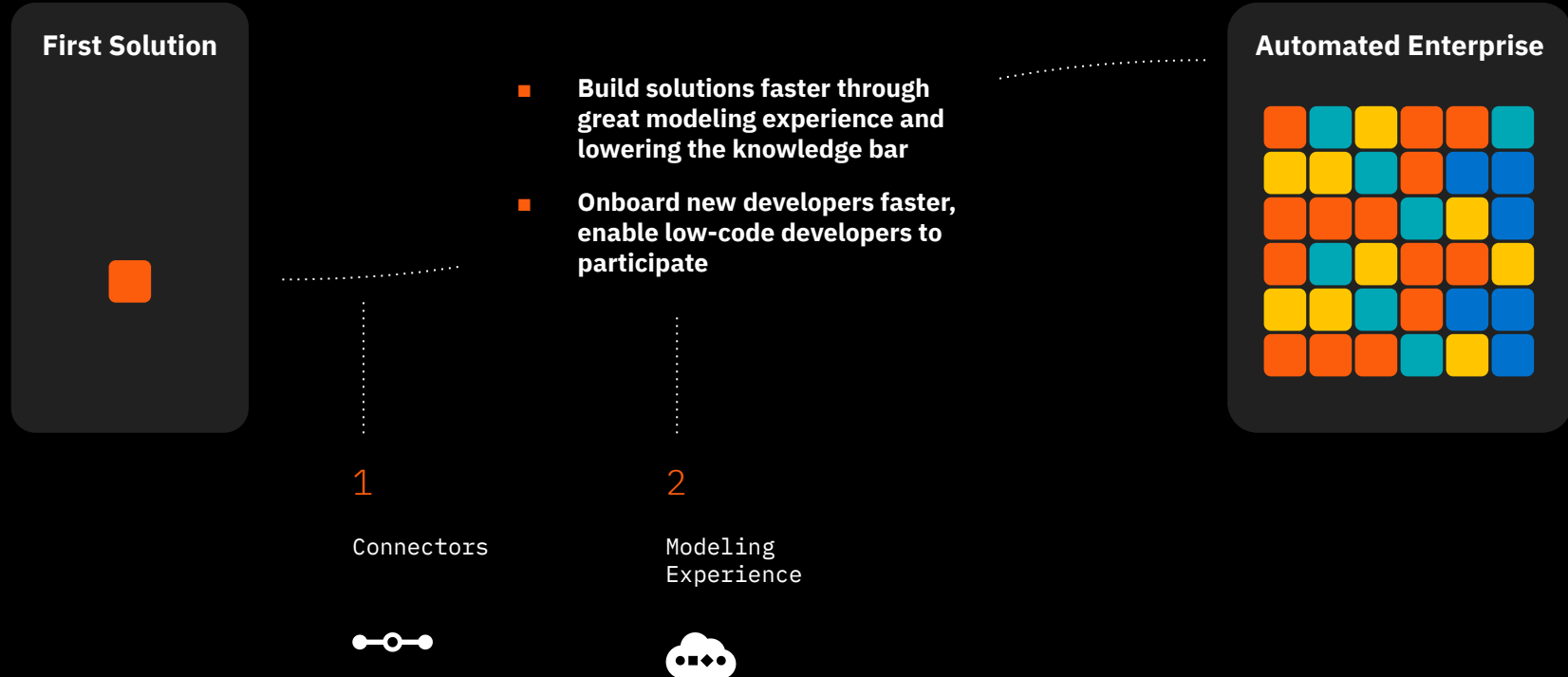
Timers 0

Child Instances 0



The Second Accelerator:

Modeling Experience



Massive Adoption in the whole Firm



Source: Presentation by Goldman Sachs at Camunda Community Summit 2022

Camunda Adoption in the Firm through our Platforms

GS Uses Camunda Platform and Camunda 8 to Provide Workflow Solutions to the Whole Firm



Zeebe Inventory						
#	CLIENT	ID	ENV	LAYOUT	VERSION	HEALTH
5	CDE	cde-uit-cluster3	UAT3	small4	1.2.4	B G E C O
6	CTRL	ctrl-dev-cluster1	DEV1	small1	1.2.4	B G E C O
7	CWM	cwm-dev-cluster2	DEV2	small1	1.2.4	B G E C O
8	CWM	cwm-dev-cluster3	DEV3	small1	1.2.4	B G E C O
9	GIR	gir-dev-cluster2	DEV2	small1	1.2.4	B G E C O
10	ICOMM	icomm-dev-cluster1	DEV1	small1	1.2.4	B G E C O
11	LEX	lex-dev-cluster1	DEV1	small1	1.2.4	B G E C O
12	LEX	lex-perf-cluster2	PERF2	large8	1.2.4	B G E C O
13	RkoMENTS	payments-dev-cluster1	DEV1	small1	1.2.4	B G E C O

300
Camunda BPMN
engines deployed

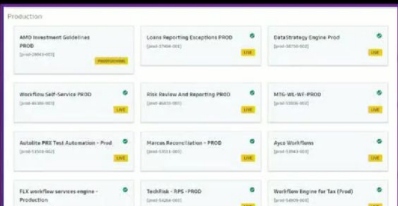
30
Camunda 8
clusters in Dev

6500
BPMN models
deployed

65K
Unique Users
per year

6 Million
Manual Tasks
weekly

15 Teams
Exploring Camunda 8
(non prod)

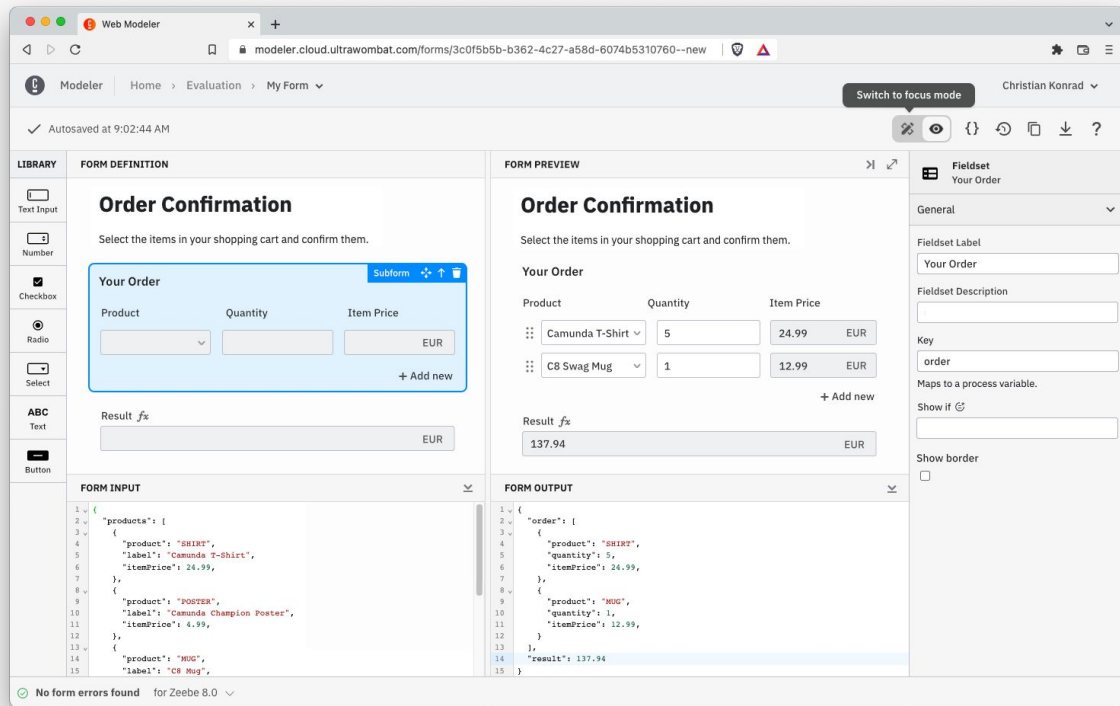


Low-code Form Builder



Coming soon:

- Interactive preview mode
- New fields and layouting mechanisms such as:
 - Flexboxes
 - Repeatable Subforms
 - Formula Fields



Meeting Employees where they are



- Slack and Microsoft Teams have established themselves as the main productivity and collaboration applications
- By bringing **human task orchestration** (e.g. the ability to start a process or complete tasks using a form) into slack, **we can reach all employees** without introducing a new tool like a tasklist
- **This lowers the bar to introduce automation to the organization**

The screenshot shows a Slack window with the Camunda app open. On the left is a sidebar with a list of users and apps. The main area displays a form titled 'Request additional hire'. The form includes the following fields:

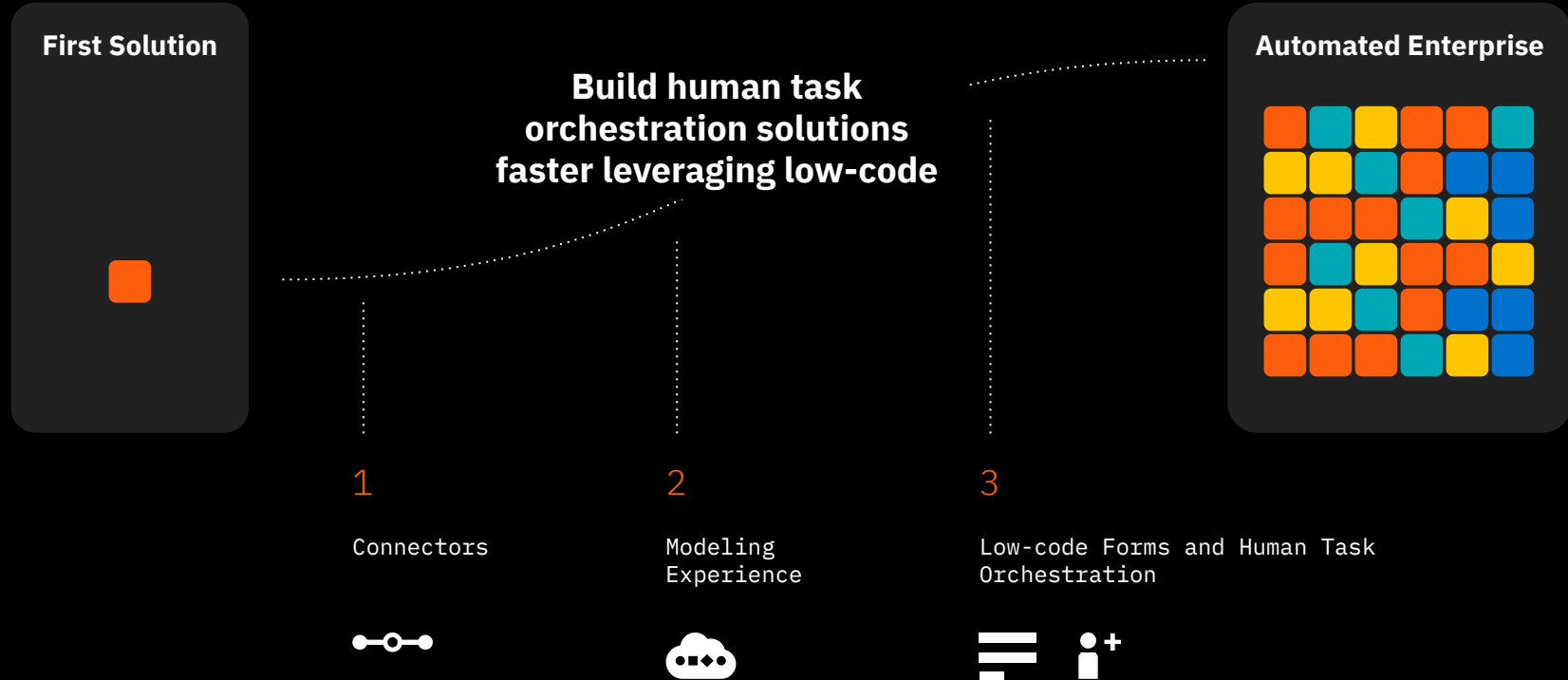
- Job Title:** A dropdown menu with the placeholder text 'Select an item'.
- Yearly salary:** A text input field with the placeholder text 'Enter the salary'.
- Yearly bonus:** A text input field containing the value '0'.
- Currency:** A dropdown menu with the placeholder text 'Select the currency for the new hire'.
- Start Date:** A date picker showing 'Today'.
- Comment:** A text area with the placeholder text 'Explain the reasoning behind the request'.

At the bottom right of the form are two buttons: 'Cancel' and 'Submit'.

Inside Camunda: Slack App as frontend of a BPMN process to request and approve new hires.

The Third Accelerator:

Low-code Forms and Human Task Orchestration



BERND



Is Camunda becoming
low-code?

Dial-in low-code as much as you need

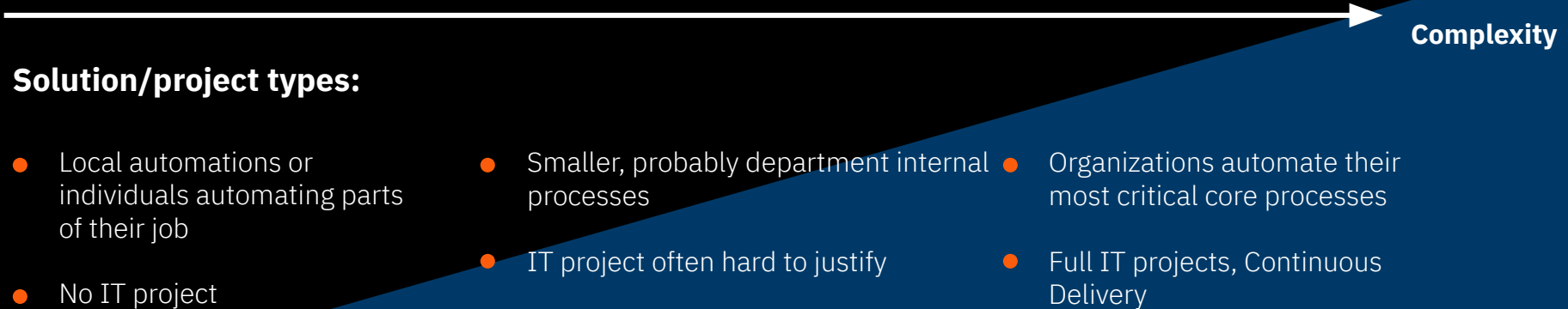


Low-code as an accelerator

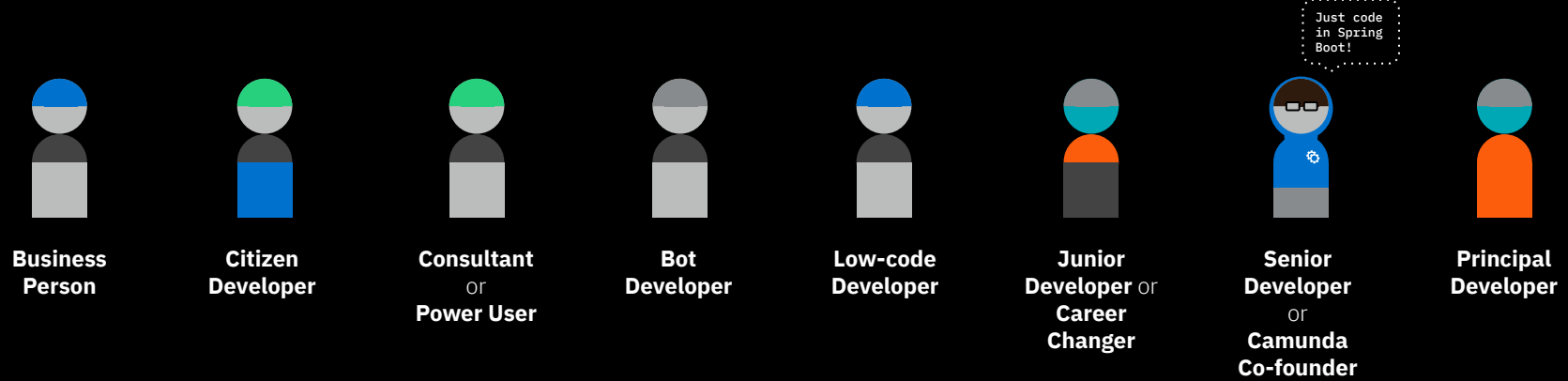


Dial-in low-code as much as you need

Project Types



Project Types and Skill Levels



Complexity

Solution/project types:

- Local automations or individuals automating parts of their job
- No IT project
- Smaller, probably department internal processes
- IT project often hard to justify
- Organizations automate their most critical core processes
- Full IT projects, Continuous Delivery

Project Types



Camunda

**Endpoint
Orchestration**
(low-code-ish)

**Bot
Orchestration**
(low-code-ish)

**Microservices
Orchestration**
(pro-code)

**Java Process
Application**
(pro-code)

Complexity

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Camunda

**Endpoint
Orchestration**
(low-code-ish)

**Bot
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**Microservices
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→ **Complexity**

Solution/project types:

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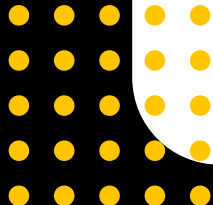
DANIEL

“

*I want teams to focus on building solutions,
not running Camunda*

Sebastian Menski

Director Camunda 8 Automation Platform Engineering, Camunda



Available as SaaS Today



The screenshot shows the 'Clusters' page in the Camunda Console. The page has a header with 'Console', 'Clusters', and 'Modeler' tabs. Below the header is a table listing clusters.

Name	Region	Generation
Weekly Webinar	Europe West	Zeebe 8.1.0-alpha3
Showroom	Europe West	Zeebe 8.0.5 - update
zeebe 8.0.6	Europe West	Zeebe 8.0.6
Camunda Internal Automation	Europe West	Zeebe 8.0.5 - update
Development	Europe West	Zeebe 8.0.5 - update

At the bottom of the page, there are links for 'Cookie Preferences' and a 'Production' button.

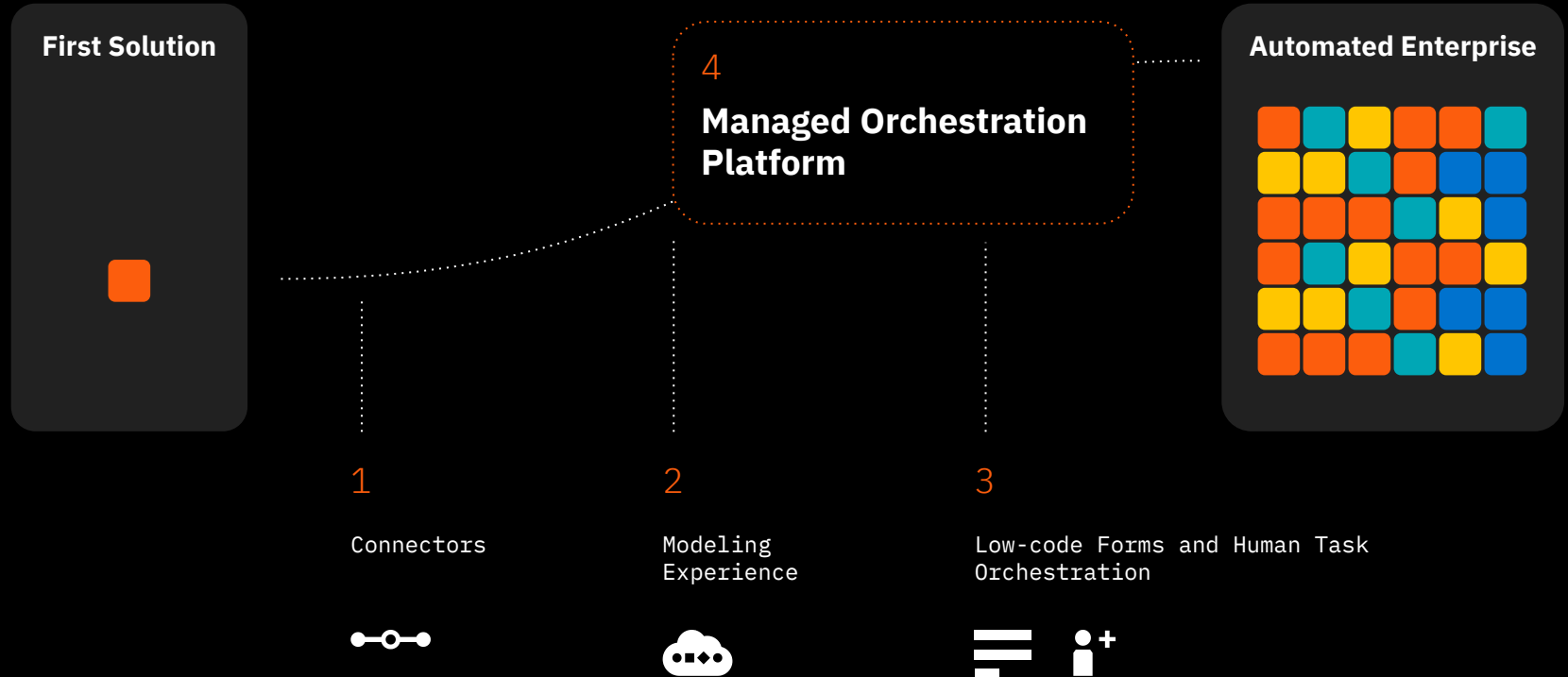
The screenshot shows the 'Clusters' page with a 'Create New Cluster' modal open. The modal has a title bar with a close button. It contains the following fields and options:

- Cluster Name** (Required): A text input field with the value 'Customer Onboarding' and a character count '19/50'.
- Channels**: Two radio buttons, 'Stable' (selected) and 'Alpha'.
- Generations**: A radio button 'Zeebe 8.0.6'.
- Region**: A radio button 'Europe West'.
- Buttons**: 'Cancel' and 'Create' buttons at the bottom right.

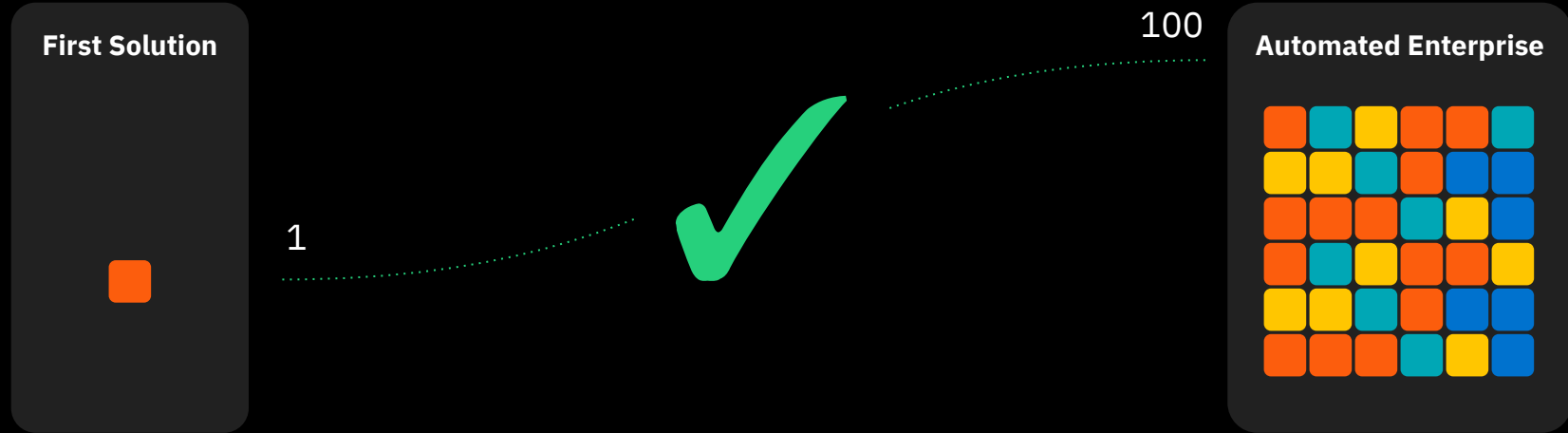
At the bottom of the page, there are links for 'Cookie Preferences' and a 'Production' button. The footer text reads: 'sha: 66ab1bc | © Camunda Services GmbH 2022. All rights reserved.'

The Forth Accelerator

Provide Camunda as a central platform to the organization

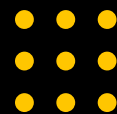


From 0 to 100 with Camunda!



Thank
you





THANK YOU



twitter.com/camunda



linkedin.com/company/camunda



camunda.com

